

# The $G_\varepsilon$ -lemma, reduced

First-order Jacobi perturbation, the pairing  $C_\varepsilon$ , and the second-order remainder  $R_2$

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## Abstract

Round 381 attacks the named remainder  $G_\varepsilon$  of [1] (the last twelve monic Jacobi coefficients  $\gamma_k$  of the  $v_2$ -chain satisfy  $|\gamma_k - \frac{1}{4}| \leq \frac{1}{16}$  and  $|\log(\gamma_{k+1}/\gamma_k)| \leq \frac{2}{5}$ ). What is proved: the first-order formula for  $\partial\gamma_k/\partial w_j$  on a fixed node set, in terms of monic orthogonal-polynomial squares and the previous norm  $h_{k-1}$  (exact over  $\mathbb{Q}$ ; floating-point finite-difference residual  $\leq 10^{-3}$  on FRAME-A  $w = 9$ ). The Jacobi-(0,1) Fejér identity and the cosine-grid mesh remain theorems of [1]. What is *not* proved: the box itself. A crude  $L^\infty$  bound on the relative arm-length perturbation is true and useless ( $\|\varepsilon\|_\infty = O(1)$ – $O(10)$  on the source, giving a first-order majorant  $\sim 3 \gg \frac{1}{16}$ ). The jump half is not a corollary of the box ( $\log \frac{5}{3} > \frac{2}{5}$ ). The second-order remainder of the weight expansion is of the same size as the first-order increment on the last twelve coefficients (midpoint test, ratio  $\geq \frac{1}{2}$ ). The hydra-legal reduction is the pair  $(C_\varepsilon, R_2)$ : the explicit first-order pairing of the weight increment against the reference OP squares, and the Taylor remainder of that expansion. A scramble world leaves  $G_\varepsilon$  (last-twelve deviation  $\geq 6.8$ ). No RH claim.

**Status.** Lemmas 1–4 are proved (finite identities). Lemma 5 is a comparison of two explicit constants. Lemma 3 is a true majorant that does not yield the box. Proposition 1 names the reduction. The hydra test of [2] is met:  $C_\varepsilon$  is a quadrature against known reference polynomials (no perturbed recurrence);  $R_2$  is a scalar remainder of a Taylor step already written. No RH claim.

## 1 Recollection

$G_\varepsilon$  of [1]: writing  $n = N - 2$  for the degree of  $v_2$ , the last twelve monic Jacobi coefficients of the bordered  $\mu - \nu$  chain satisfy

$$|\gamma_k - \frac{1}{4}| \leq \frac{1}{16}, \quad |\log(\gamma_{k+1}/\gamma_k)| \leq \frac{2}{5}, \quad k \in [n - 12, n). \quad (1)$$

The mesh  $x = \cos(2\pi k/L)$ ,  $L = 4h - 2$ , is a construction identity. Equal weights on the occupied cosine nodes are Chebyshev ( $\gamma_1 = \frac{1}{2}$ ,  $\gamma_k = \frac{1}{4}$  for  $k \geq 2$  up to a  $10^{-5}$  residual from one missing fold). The Fejér factor  $2(1 - x)$  is Jacobi-(0,1) with  $|\gamma_n - \frac{1}{4}| = 1/(4(2n + 1)^2)$ . The source formula of [1] is  $\gamma_k = \frac{1}{4} + \delta_k^{\text{Fejér}} + \delta_k^{d_{\text{arm}}}$ , and  $\delta^{\text{Fejér}}$  is negligible at half-filling on the *unsigned* occupied grid. The live channel is the arm-length weight  $d_{\text{arm}}$  (r342 digamma  $F_A(\xi) = -\log \pi + \Re \psi(\frac{1}{4} + i\xi/2)$  plus the two-cell tent of the prime layer). On the 181-window surface [1] the box holds (max last-12 0.0519, max jump 0.337). This note does not repeat that census.

## 2 The first-order formula

Let  $\sigma = \sum_j w_j \delta_{x_j}$  be a quasi-definite atomic measure on a fixed node set (signed weights allowed). Write  $p_k$  for the monic orthogonal polynomials,  $h_k = \int p_k^2 d\sigma$ , and  $\gamma_k = h_k/h_{k-1}$ .

**Lemma 1** (first-order Jacobi). *At a quasi-definite weight,*

$$\frac{\partial \gamma_k}{\partial w_j} = \frac{p_k(x_j)^2 - \gamma_k p_{k-1}(x_j)^2}{h_{k-1}}.$$

*Equivalently, for a multiplicative increment  $dw_j = w_j \varepsilon_j$ ,*

$$\delta \gamma_k^{\text{FO}} = \gamma_k \left( \langle \varepsilon p_k^2 \rangle_\sigma / h_k - \langle \varepsilon p_{k-1}^2 \rangle_\sigma / h_{k-1} \right).$$

*For  $\sigma = \mu - \nu$  the  $\nu$ -atoms enter with the opposite sign.*

*Proof.* The leading coefficient of  $p_k$  is identically 1, so  $\partial p_k / \partial w_j$  has degree at most  $k-1$ . Orthogonality gives  $\int p_k (\partial p_k / \partial w_j) d\sigma = 0$ . Differentiating  $h_k = \sum_i w_i p_k(x_i)^2$  therefore yields  $\partial h_k / \partial w_j = p_k(x_j)^2$ , with no contribution from the polynomial variation. The quotient rule on  $\gamma_k = h_k/h_{k-1}$  is the displayed formula.  $\square$

The identity is gated over  $\mathbb{Q}$  on a six-atom rational toy (central difference versus the formula, relative residual  $< 3 \cdot 10^{-4}$  at step 1/200) and on FRAME-A  $w = 9$  (one  $\mu$ -atom,  $dw = 10^{-7}$ , relative residual  $\leq 10^{-3}$  through degree 12).

The first-order remainder of a finite step  $\varepsilon$  is quadratic. On the same toy,  $|\Delta\gamma - \text{FO} \cdot \varepsilon|/\varepsilon^2$  lies in  $[0.05, 5]$ . That is Lemma 2.

**Lemma 2** (quadratic remainder, toy). *On the six-atom rational measure, the first-order Taylor remainder in a single-weight step of size  $\varepsilon = 1/200$  satisfies  $0.05 \leq |R_2|/\varepsilon^2 \leq 5$ .*

## 3 What the $L^\infty$ bound cannot do

Write  $\varepsilon_j = (w_j/\bar{w}_j) - 1$  for the Fejér-stripped relative arm-length,  $\bar{w} \propto (1-x)$ .

**Lemma 3** (crude majorant).  $|\delta \gamma_k^{\text{FO}}| \leq \gamma_k (\|\varepsilon\|_\infty^{(\mu)} + \|\varepsilon\|_\infty^{(\nu)})$  up to the signed-mass normalisation of the orthonormal squares. On FRAME-A  $w = 9$  one has  $\|\varepsilon_\mu\|_\infty > 1$ , and the resulting majorant is  $\sim 3 \gg \frac{1}{16}$ .

The bound is true and does not prove (1). The measured last-twelve deviation is 0.034; the majorant overshoots by two orders of magnitude because  $\langle \varepsilon, \pi_k^2 - \pi_{k-1}^2 \rangle$  cancels. That pairing against the *reference* OP squares (the  $\mu$ -chain, or the Fejér-constant reference) is the object  $C_\varepsilon$  below. It does not require solving the perturbed recurrence.

A PNT-free bound  $\Lambda(n) \leq \log n$  is elementary (trial division:  $\Lambda$  vanishes off prime powers, and  $\log p \leq \log n$  on  $n = p^k$ ).

**Lemma 4** (von Mangoldt). *For every integer  $n \geq 2$ ,  $\Lambda(n) \leq \log n$ .*

This bounds cell-masses of the r342 tent, hence  $|c_q^{\text{atom}}|$  in each lag cell. It does *not* bound the cancelled pairing: high-lag  $|c_q|$  on MAIN and on a scramble of  $w = 9$  are of the same order, while last-twelve  $\gamma$  is not.

## 4 The jump is independent

**Lemma 5** (jump not implied). *The closed box  $|\gamma - \frac{1}{4}| \leq \frac{1}{16}$  permits the consecutive ratio  $5/3$ , and  $\log \frac{5}{3} > \frac{2}{5}$ .*

The jump half of  $G_\varepsilon$  is a Lipschitz statement on consecutive last-twelve coefficients, not a corollary of the uniform box. On the source it is cheaper (FRAME-A  $w = 9$ : jump  $0.249 < 0.4$ ;  $\mu$ -only jump  $0.122$ ). The same pairing that controls  $C_\varepsilon$  at  $k$  and at  $k+1$  controls the difference; the named remainder below includes it.

## 5 The smaller lemmas

Take as reference the *positive*  $\mu$ -chain on the  $+x$  arm (same nodes, unsigned). Write  $\gamma_k^\mu$  for its monic coefficients, and let  $\text{FO}_k[-\nu]$  be Lemma 1 applied to the finite increment that subtracts  $\nu$ . Define

$$C_{\varepsilon,k} := |\text{FO}_k[-\nu]|, \quad (2)$$

$$R_{2,k} := \gamma_k(\mu - \nu) - \gamma_k^\mu - \text{FO}_k[-\nu]. \quad (3)$$

**Definition 1** ( $C_\varepsilon$  and  $R_2$ ). *Condition  $C_\varepsilon$  is  $C_{\varepsilon,k} \leq \frac{1}{32}$  for  $k \in [n-12, n)$ . Condition  $R_2$  is  $|R_{2,k}| \leq \frac{1}{32}$  on the same range. Condition  $G_\varepsilon^\mu$  is (1) for the positive  $\mu$ -chain (box  $\frac{1}{32}$  and jump  $\frac{1}{5}$  suffice).*

**Proposition 1** (reduction).  *$C_\varepsilon$  and  $R_2$  are strictly more elementary than  $G_\varepsilon$  in the sense of [2] §5:*

1. Finite. *Twelve real inequalities on an explicit sum against the unperturbed OP squares ( $C_\varepsilon$ ), plus twelve scalar remainders ( $R_2$ ).*
2. Closer to the construction. *No Prüfer phase, no mask remainder, no  $v_2$  occupancy.  $C_\varepsilon$  does not even require the perturbed three-term recurrence.*
3. Fewer quantifiers. *No  $\forall\eta$ .*

On FRAME-A  $w = 9$ :  $G_\varepsilon$  holds (last-12 0.034, jump 0.249);  $G_\varepsilon^\mu$  holds (last-12 0.026, jump 0.122);  $\text{mass}(\nu)/\text{mass}(\mu) = 0.061$ ; the midpoint test of  $R_2$  along the Fejér-constant path at degree 40 gives  $|\text{quad}|/|\text{lin}| \geq \frac{1}{2}$  on the last twelve —  $R_2$  is not absorbed by the first-order term. The chain

$$T'_2 \Leftarrow g_i \geq \frac{3}{8} \Leftarrow \text{MED-CAP} \Leftarrow X_n \Leftarrow V_2 \Leftarrow V'_3 \Leftarrow G_\varepsilon \Leftarrow G_\varepsilon^\mu \wedge C_\varepsilon \wedge R_2$$

stands, with  $(C_\varepsilon, R_2)$  the remaining named pair ( $G_\varepsilon^\mu$  is the positive-measure sibling: same box, no signed Stieltjes).

The expansion  $\mu \rightarrow \mu - \nu$  is a *finite*  $O(1)$  step in the Christoffel sampling  $\langle \pi_k^2 \rangle_\nu$  (measured last-twelve maximum 0.389 on FRAME-A  $w = 9$ ). First-order perturbation theory about  $\mu$  is therefore not a small- $\varepsilon$  expansion, which is why  $R_2$  is comparable to the linear increment. Expanding about the actual source against a  $10^{-4}$  dictionary error would make  $R_2$  tiny and would move the whole content of  $G_\varepsilon$  onto the Jacobi coefficients of the r342 dictionary measure — the same lemma, not a smaller one.

## 6 Adversaries

**Segregated block.** The  $V'_3$  obstruction of [1] is a slow-then-fast *step* block. Lemma 1 is a weight identity and is not refuted by a step rearrangement. A low-mode weight cluster (the natural weight analogue of spatial segregation) does not move the last twelve  $\gamma_k$ : on an equal-weight cosine grid of length 80, a relative amplitude 0.4 in  $\cos 2\theta$  changes last-twelve  $\gamma$  by  $4 \cdot 10^{-5}$ . The block sits outside the high-lag tent class that can move the last twelve, and is not an FO-counterexample.

**Scramble.** FRAME-A  $w = 9$ , scramble seed 1: last-twelve  $|\gamma - \frac{1}{4}| \geq 6.8$ , jump  $\gg \frac{2}{5}$ . The lag-coefficient  $L^\infty$  of  $d_{\text{arm}}$  is *not* the separator (high-lag  $|c_q|$  is comparable to MAIN). What explodes is the  $\mu$ -Christoffel sampling of  $\nu$ :  $\langle \pi_k^2 \rangle_\nu$  stays in  $(0.34, 0.40)$  on MAIN last-twelve and exceeds  $10^3$  on the scramble. That is the named source bound scramble breaks, and it is part of  $C_\varepsilon$  (the pairing evaluates  $\pi_k$  of  $\mu$  at the  $\nu$  nodes).

**Second order.** Along the path from Fejér-constant  $d_{\text{arm}}$  to the actual weights, at degree 40, the last-twelve midpoint remainder is of the same size as the linear increment (ratio  $\geq \frac{1}{2}$ ). At full depth  $n = N - 2$  the quadratic increment is larger.  $R_2$  is not dominated. The reduction to  $R_2$  is forced by Leg D(iii), not optional.

## 7 What remains

To prove  $G_\varepsilon$ : prove  $C_\varepsilon$  (the cancelled pairing of  $d_{\text{arm}}$  against  $\pi_k^2 - \pi_{k-1}^2$  of the reference, uniformly in the last twelve  $k$ ) and  $R_2$  (the Taylor remainder of the same finite step), plus  $G_\varepsilon^\mu$  if the reference is the  $\mu$ -chain. The Geronimus/Fourier picture — last-twelve  $\delta\gamma$  tracks the high-lag tent, i.e. Fourier mode  $q \sim M$  of  $d_{\text{arm}}$  — is the natural attack on  $C_\varepsilon$ , and is not carried here. T1 / COMPOSE unchanged. No RH claim.

## 8 Machine checks

Numbered lemmas: `verify_g_eps.py` (same directory). Standalone: FO formula over  $\mathbb{Q}$ ; quadratic remainder; Jacobi-(0,1); jump comparison;  $\Lambda(n) \leq \log n$ ; Chebyshev quarters. Construction: FO versus finite difference,  $G_\varepsilon$  and  $G_\varepsilon^\mu$  on FRAME-A  $w = 9$ ,  $R_2$  midpoint ratio, scramble kill,  $L^\infty$  crude, two-period versus block. Discovery census: `g_eps_lemma_probe.py` (`--smoke` / full). Exit line: `G_EPS LEMMA VERIFIED`.

## Claim boundary

Finite identities for Jacobi perturbation on a fixed node set, a named reduction of  $G_\varepsilon$  to  $(C_\varepsilon, R_2)$ , and a scramble break of the source pairing. No statement about zeros of  $\zeta(s)$ , in either direction. No RH claim.

## References

- [1] S. Hamann, *The  $V'_3$ -lemma, reduced*, standalone note, August 2026, `rh/problem/v3prime_proof.tex`.
- [2] S. Hamann, *Run-length regularity  $V_2$  at the  $x$ -mask*, standalone note, August 2026, `rh/problem/v2_regularity.tex`.